# *Modeling and coding of spot microphone signals for immersive audio based on the sinusoidal model*

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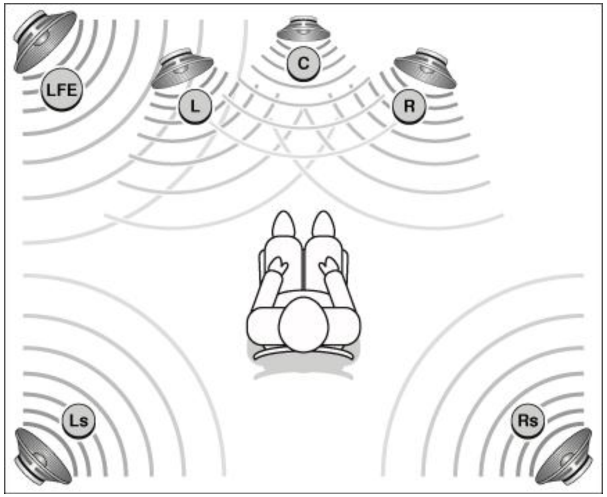
# *Outline*

- Introduction
- Scope of the proposed method
- Background
  - sinusoids plus noise model
- Analysis of the proposed model
  - coding of the modeling parameters
- Experimental results and demo
- Conclusions

# *Introduction (1/2)*

- Multichannel audio surrounds the listener with a realistic acoustic scene




- Increased demand on high datarates because of the large number of channels

- Exploit inter- and intra- channel redundancy to reduce coding rate (MPEG AAC, Dolby AC-3, etc.) with bitrates around 320 kbps for 5.1

- ***Spatial audio coding (SAC)*** *was* recently proposed for more actively exploiting inter-channel redundancy with bitrates around 64 kbps for 5.1

# *Introduction (2/2)*

- ***Immersive audio systems:*** immerse the listener into a virtual acoustic scene
- Immersive audio is based on enhanced audio content
  - as many sound sources as possible
  - interactivity between the user and the audio environment
- Immersive audio applications:
  - geographically distributed musicians' collaboration
  - tele-presence in a concert hall performance
- Recordings for multichannel and immersive audio contain a higher number of microphones than the available loudspeakers
  - these are called ***spot microphone signals***
  - a mixing process is needed
  - ***interactivity: remote mixing*** → dynamically changing content

# *Scope of the proposed method*

- Transmit the channels before the mixing process
  - encode each spot signal before mixing
  - allow for the individual spot signals to be available at the decoder for remote mixing
- The proposed method is based on:
  - full encoding of one channel only (reference channel, can be a downmix of the spot signals or one of the signals)
  - model each spot signal and decode using the reference channel
- High-quality low bitrate of spot signals (in the order of 20 kbps/spot signal)
- Apply the sinusoids-plus-noise model for obtaining the per spot signal side information

# Background information

## Sinusoids plus noise model

- Audio signal is modeled as: $$s(n) = \underbrace{\sum_{l=1}^{L} A_l(n)\cos(\theta_l(n))}_{\text{deterministic component}} + \underbrace{e(n)}_{\text{noise component}}$$
  - ***L***: number of sinusoids
  - ***$A_l$(n)***: instantaneous amplitude
  - ***$\theta_l$(n)***: instantaneous phase
  - ***e(n)***: modeling error

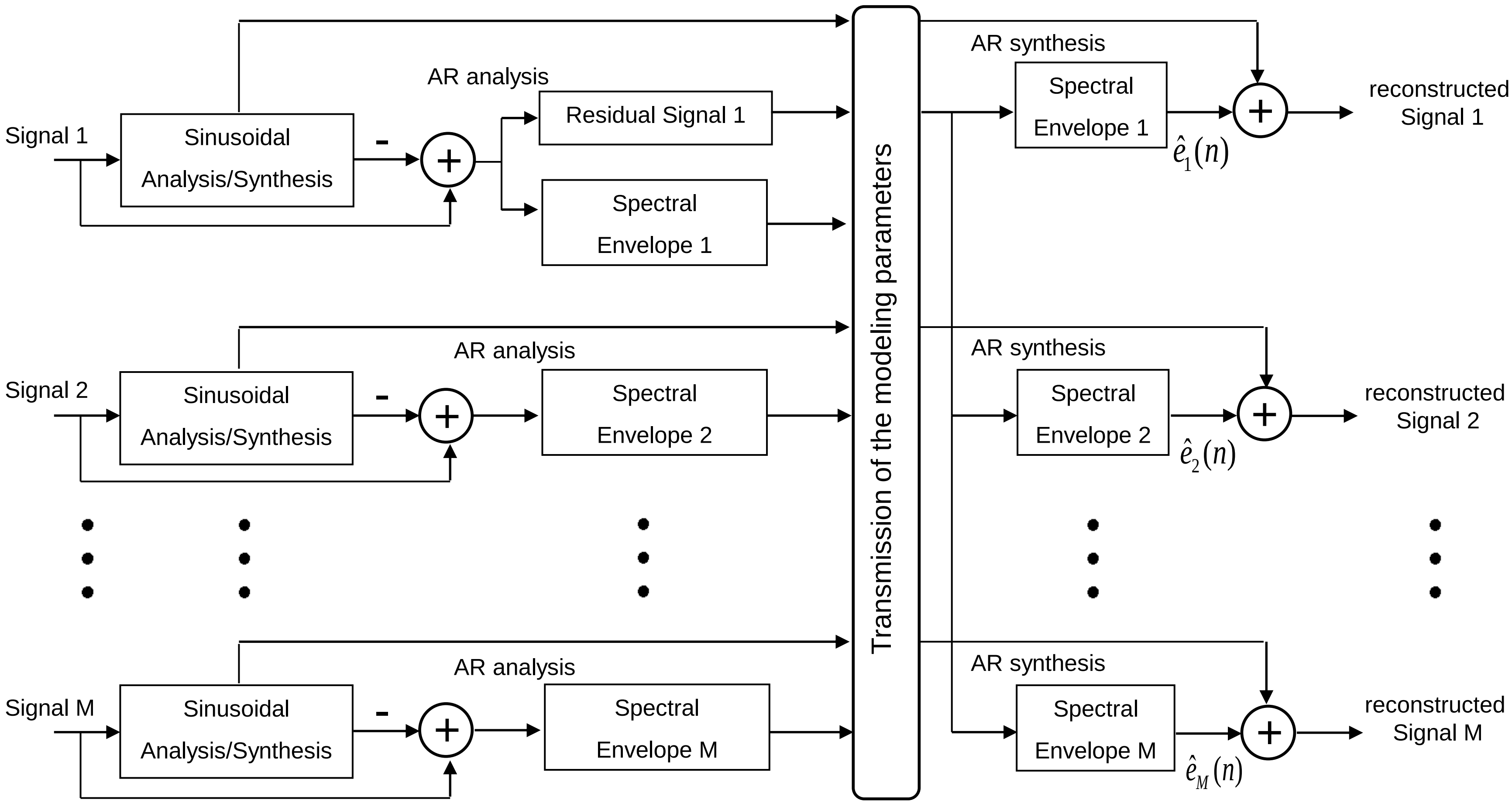
- Use Linear Prediction (LP) analysis to model the noise spectral envelope

$$e(n) = \sum_{i=1}^{p} a(i)e(n-i) + r_e(n) \quad \overset{frequency \;\; domain}{\Rightarrow} \quad S_e(e^{j\omega}) = \left| \frac{1}{A(e^{j\omega})} \right|^2 S_{r_e}(e^{j\omega})$$

  - ***$r_e$(n)***: residual of the noise *e(n)*
  - ***p***: AR filter order
  - *p+1$^{th}$-dimension vector* α represents the spectral envelope of *e(n)*

# Proposed method

- Consider a recording with *M* microphone spot signals
  - signal 1 is the reference signal



# *Coding of the modeling parameters*

- The coding process is divided into:
  1. coding of the sinusoidal parameters
  2. coding of the noise spectral envelopes

- The sinusoidal parameters (amplitude, frequency, phase) are jointly quantized using a high-rate quantization scheme [1]

- The LPC coefficients of the spectral envelopes are quantized using a vector quantization scheme [2]

[1] R. Vafin et.al., "On Frequency Quantization in Sinusoidal Audio Coding", *IEEE Signal Proc. Letters*

[2] A. D. Subramaniam et.al., "PDF optimized parametric vector quantization of speech line spectral frequencies", *IEEE Trans. on Speech and Audio Proc.*

# *Experimental results*

*Performance evaluation of modeling*

- Start from two different monophonic microphone signals
  - reference channel is the downmix of the two signals
- Test our method in a stereophonic setting using headphones
- Create stereophonic signals by mixing two monophonic signals
  - amplitude panning (±14 dB relative level difference)
  - anchor signals: distorted image width (±2.5 dB relative level difference)
- Compare stereo mix of original spot signals with the stereo mix of the modeled signals

# *Experimental results*

- Created the following test signals

| |
|---|
| 1. bass singer plus soprano singer<br>2. guitar plus rock singer<br>3. harpsichord plus violin<br>4. female plus male speech<br>5. trumpet plus violin<br>6. violin plus guitar<br>7-9. male chorus plus female chorus |

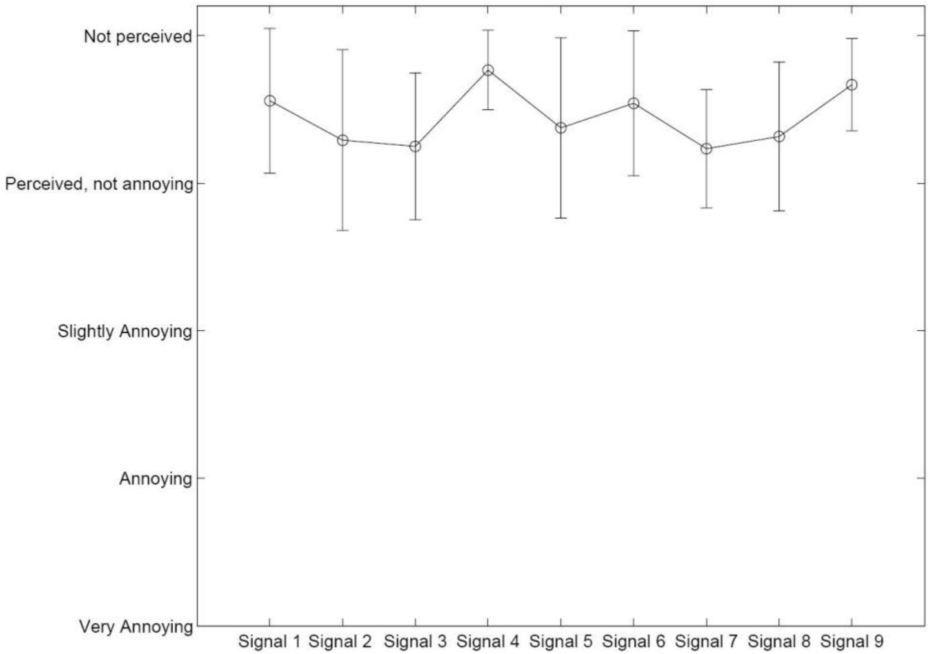
- ❑10 sinusoids per audio frame
- ❑30 msec. sinusoidal analysis/synthesis frame
- ❑50% overlapping
- ❑LP order for the noise shaping filter is 10

- Listening tests to examine the quality and image width distortion
  - 11 listeners participated

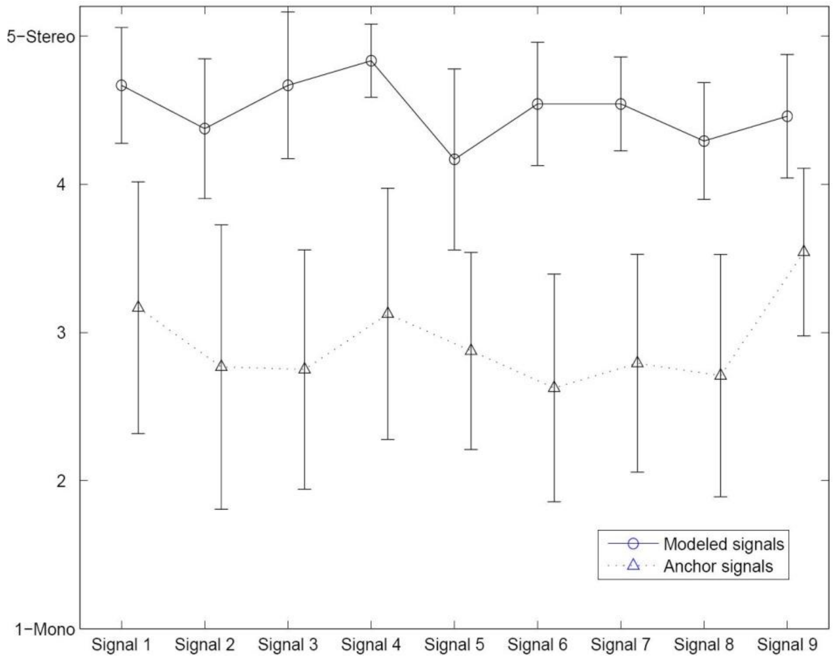
# Experimental results

## *Performance evaluation of modeling*

Evaluate quality




Evaluate image width



# Experimental results

*Performance evaluation of coding*

- Used Signals 7-9 of the previous test
  - Concert hall recordings, most complex of the testing set
- Experimental parameters:
  - 30 msec. analysis/synthesis
  - 50% overlapping
  - LP order for the noise shaping filter is 10
  - 23 msec. frame for noise part with 75% overlapping

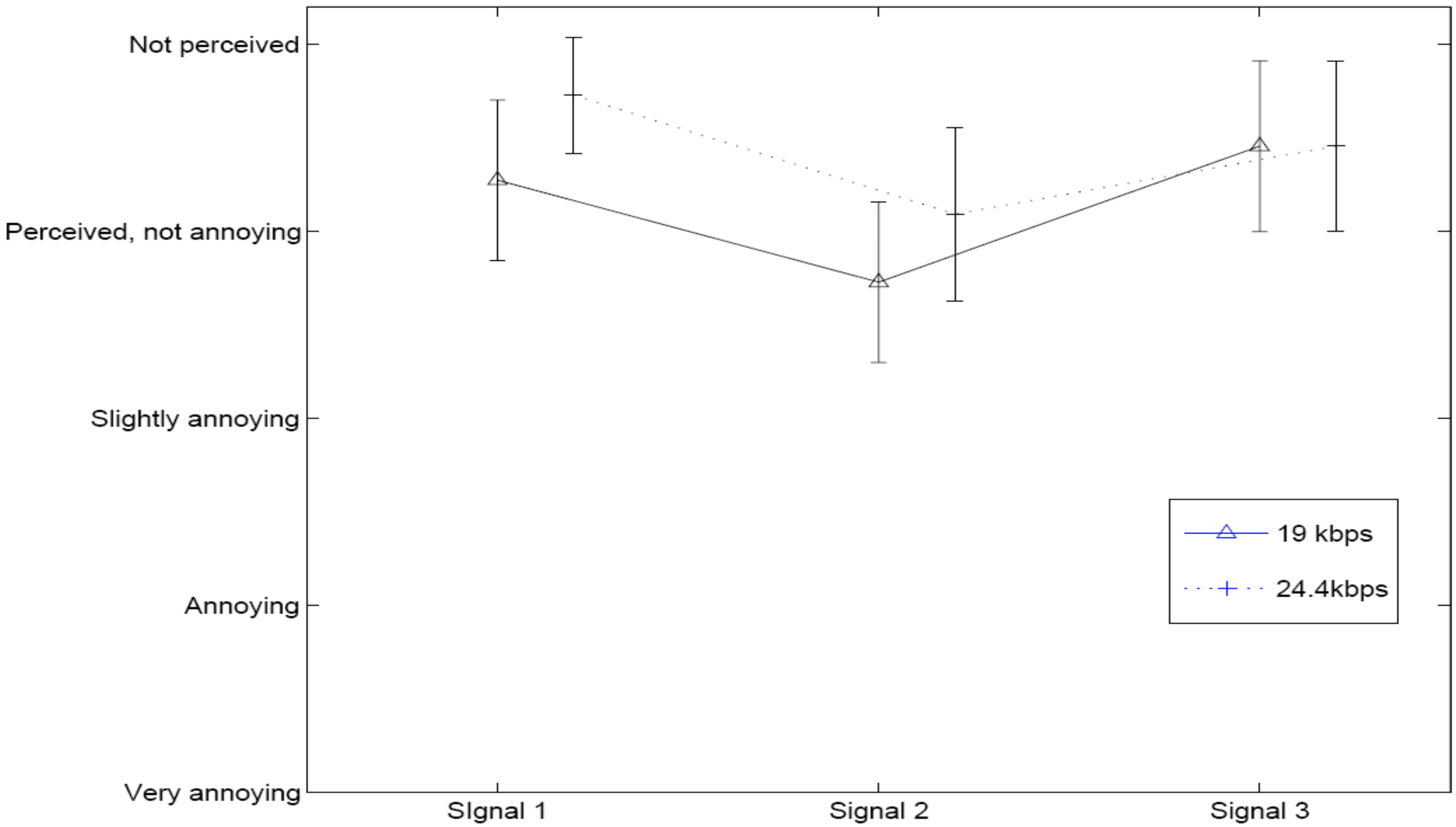
10 sinusoids: 28 bits (19.6 kbps)
20 bits (14.2 kbps)

28 bits (4.8 kbps)

- DCR-based listening tests to compare the quality of the coded signal vs. original
  - 11 listeners participated
  - used three 10 sec parts from the two performances (signals 1-3)

# Experimental results

## *Performance evaluation of coding*



# *Modeling Demos*

- Demo1: Bass and Soprano Example
  - Original stereo mix
  - Modeled stereo mix (10 sinusoids, downmix reference)
- Demo 2: Male Chorus and Female Chorus
  - Original stereo mix
  - Modeled stereo mix (10 sinusoids, downmix reference)
- All sound files used in the tests can be found at

  http://www.ics.forth.gr/~mouchtar/snm/

# *Conclusions*

- A sinusoids plus noise model was presented tailored for modeling spot signals
- Take advantage of interchannel similarities
- Allow for remote applications based on the availability of the spot signals at the decoder side
- The perceived quality remains high even for a low number of the sinusoidal parameters
  - low bitrate → 19 kbps per spot signal

Thank You!!